

❖ AMI – Amazon Machine Image

❖ What is an AMI?

- An Amazon Machine Image (AMI) is a template that contains the software configuration required to launch an EC2 instance.

It includes:

1. **Operating system** (e.g., Amazon Linux, Ubuntu, Windows)
2. **Pre-installed software** (web servers, databases, custom applications)
3. **Settings and permissions**
4. **Root volume and data volumes**

When you launch an instance, you select an AMI to define the initial state of the machine

❖ Types of AMIs

- Amazon provides multiple types of AMIs depending on the use case:

Type	Description
AWS-provided AMIs	Official images provided and maintained by AWS (e.g. Amazon Linux, Windows Server)
Marketplace AMIs	Third-party AMIs available for purchase or subscription in AWS Marketplace
Community AMIs	AMIs shared publicly by AWS users (may not be verified or supported)
Custom AMIs	User-created AMIs built from an existing EC2 instance

❖ What is a Launch Template?

- A Launch Template in Amazon EC2 is a reusable configuration for launching EC2 instances.
It **defines** the instance settings such as **AMI ID, instance type, key pair, security groups, storage configuration, tags, and network interfaces**

❖ Why Use Launch Templates?

- Launch Templates in AWS simplify and standardize the process of launching EC2 instances by saving all the configuration details in one reusable template.

- Key Benefits:

- i. Reusability
- ii. Version Control
- iii. Integration with Other AWS Services
- iv. Parameter Consistency
- v. Overrides Supported
- vi. Automation Friendly

❖ How do you migrate an EC2 instance from one region to another?

1. Create an Amazon Machine Image (AMI) of the existing instance.
2. Copy the AMI to the target region.
3. Launch a new EC2 instance from the copied AMI in the target region